

LoRA: Low-Rank Adaptation of Large Language Models

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An important paradigm of natural language processing consists of large-scale pre-training on general domain data and adaptation to specific tasks. As models grow larger, full fine-tuning becomes prohibitive. We propose Low-Rank Adaptation (LoRA), which freezes the pre-trained model weights and injects trainable rank decomposition matrices into each layer of the Transformer architecture.

Introduction

Many applications in natural language processing rely on adapting one large, pre-trained language model to multiple downstream applications. In full fine-tuning, all model parameters are updated, creating a distinct copy of the model for every task. For a 175B GPT-3 model, storing and deploying fine-tuned models is extraordinarily expensive.

Methodology

We propose Low-Rank Adaptation (LoRA). We hypothesize that the parameter updates ΔW during task adaptation have a low 'intrinsic dimension'. For a pre-trained weight matrix W_0 in $\mathbb{R}^{d \times k}$, we constrain its update by representing $\Delta W = B * A$, where B in $\mathbb{R}^{d \times r}$ and A in $\mathbb{R}^{r \times k}$, with $\text{rank } r \ll \min(d, k)$. During training, W_0 is frozen and does not receive gradient updates, while A and B contain trainable parameters.

Experiments & Evaluation

We evaluate LoRA on RoBERTa, DeBERTa, and GPT-2, scaling up to GPT-3 175B. Tasks include GLUE benchmark, E2E NLG Challenge, and WebNLG. We set rank r to 4 or 8 across self-attention projection weights.

Results & Discussion

LoRA can reduce the number of trainable parameters by 10,000 times and GPU memory consumption by 3 times compared to full fine-tuning. LoRA performs on par or better than full fine-tuning on GLUE with RoBERTa and GPT-3 175B, while introducing zero additional inference latency.

Limitations

LoRA is primarily evaluated on text-to-text generative and classification benchmarks. It does not compress the underlying pre-trained base model weights in memory during inference if serving multiple distinct base models concurrently.

Conclusion & Future Work

LoRA provides a parameter-efficient, modular adaptation framework for foundation models. Future work includes investigating combinations of LoRA with quantization (QLoRA) and dynamic rank allocation.